

Ex 9.4 Q40

$$\sum_{n=1}^{\infty} \frac{2^n + 3^n}{3^n + 4^n}$$

$2^n, 3^n, 4^n \in \{c^n \mid c > 1\}$ (Exponential functions)

Only one type $\rightarrow$ LCT

$$\left. \begin{array}{l} 2^n + 3^n \sim 3^n \\ 3^n + 4^n \sim 4^n \end{array} \right\} \frac{2^n + 3^n}{3^n + 4^n} \sim \frac{3^n}{4^n} = \left(\frac{3}{4}\right)^n$$

Geometric series

Let $a_n = \frac{2^n + 3^n}{3^n + 4^n}$ and $b_n = \left(\frac{3}{4}\right)^n$ for $n \geq 1$.

$$\begin{aligned} \lim_{n \rightarrow \infty} \frac{a_n}{b_n} &= \lim_{n \rightarrow \infty} a_n \cdot \frac{1}{b_n} = \lim_{n \rightarrow \infty} \frac{(2^n + 3^n)}{(3^n + 4^n)} \cdot \frac{4^n}{3^n} \times \frac{1}{\frac{1}{4^n}} \\ &= \lim_{n \rightarrow \infty} \frac{(2^n + 3^n)}{\left[\left(\frac{3}{4}\right)^n + 1\right]} \cdot \frac{1}{\frac{1}{3^n}} \times \frac{1}{\frac{1}{3^n}} \\ &= \lim_{n \rightarrow \infty} \frac{\left(\frac{2}{3}\right)^n + 1}{\left(\frac{3}{4}\right)^n + 1} \quad \because \left(\frac{2}{3}\right)^n \rightarrow 0 \\ &= 1 > 0 \quad \left(\frac{3}{4}\right)^n \rightarrow 0 \end{aligned}$$

Since $\sum_{n=1}^{\infty} b_n = \sum_{n=1}^{\infty} \left(\frac{3}{4}\right)^n$ converges, $\rightarrow$ because $\frac{3}{4} < 1$
(geometric series)

$\sum_{n=1}^{\infty} a_n$ converges by LCT.

Q 27 $\sum_{n=3}^{\infty} \frac{1}{\ln(\ln n)}$

Identify functions $\rightarrow$ It is not only one
a standard form
(LCT fails)

$\ln x \ll x$ for $x \gg 1$

$\ln n \ll n^c$
($c > 0$)

$\ln n \leq n$ for $n \gg 1$

$\ln x < 0$ when $x < 1$

$0 \leq \ln(\ln n) \leq \ln(n) \rightarrow \frac{1}{n} \leq \frac{1}{\ln n}$

$\frac{1}{n} \leq \frac{1}{\ln n} \leq \frac{1}{\ln(\ln n)}$ for $n \gg 3$

Since $\sum_{n=3}^{\infty} \frac{1}{n}$ is divergent, $\sum_{n=3}^{\infty} \frac{1}{\ln(\ln n)}$ is ~~not~~ divergent by DCT.